

Parkinsons Work

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```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.1      ✓ readr      2.1.4
## ✓ forcats    1.0.0      ✓ stringr    1.5.0
## ✓ ggplot2    3.4.2      ✓ tibble     3.2.1
## ✓ lubridate  1.9.2      ✓ tidyr      1.3.0
## ✓ purrr      1.0.1
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(randomForest)
```

```
## randomForest 4.7-1.1
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##      combine
##
## The following object is masked from 'package:ggplot2':
##
##      margin
```

```
library(rfExtra)
```

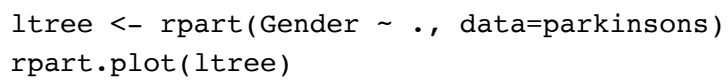
```
##  
## Attaching package: 'rfExtra'  
##  
## The following objects are masked from 'package:randomForest':  
##  
##      combine, randomForest  
##  
## The following object is masked from 'package:dplyr':  
##  
##      combine
```

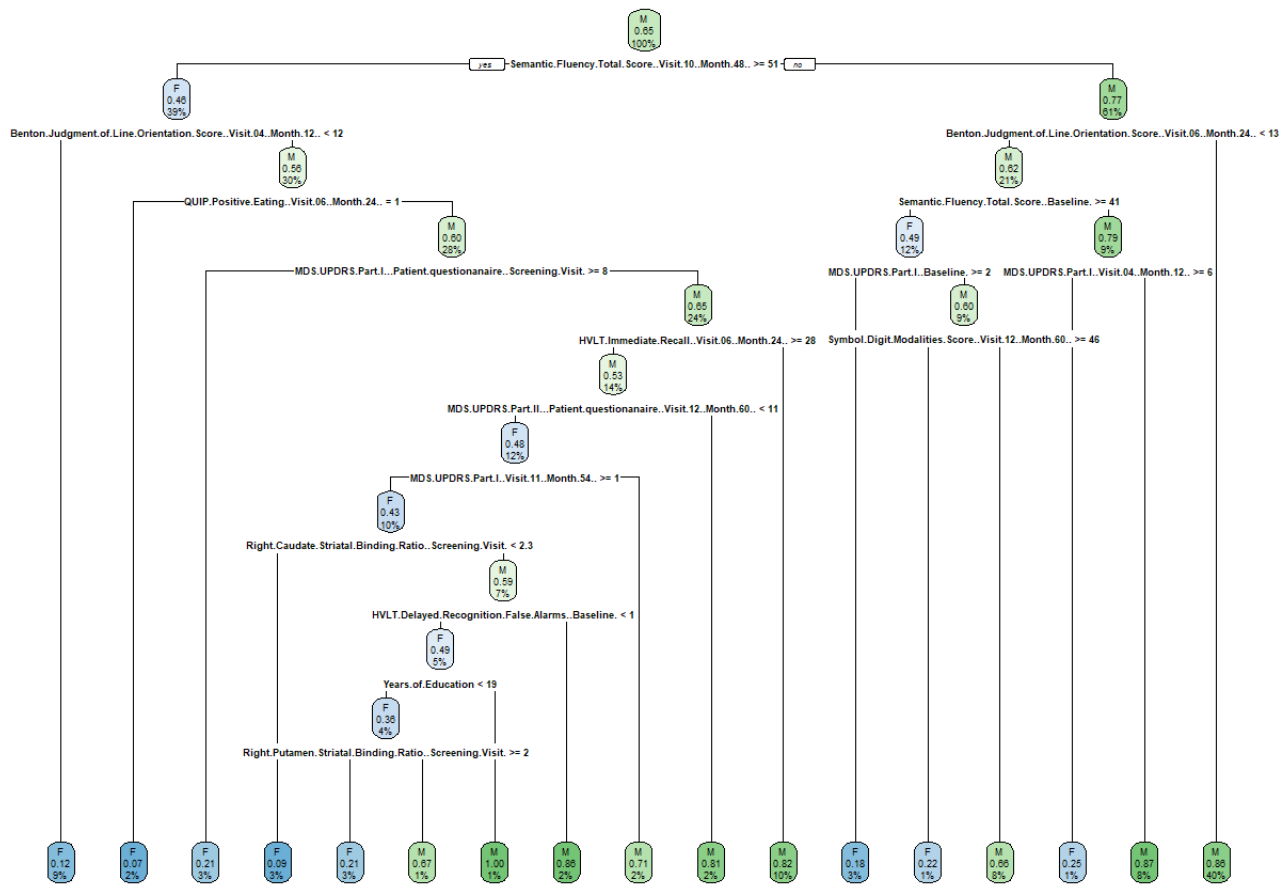
```
library(parallel)  
library(rpart)  
library(rpart.plot)
```

```
#parkinsons = read.csv('ppmi_data.csv',check.names = FALSE, stringAsFactors = TRUE)  
  
parkinsons <- read_csv("ppmi_data.csv", show_col_types = FALSE)  
  
parkinsons = data.frame(parkinsons,stringsAsFactors = TRUE)  
  
#parkinsons <- parkinsons %>%  
  # rename(Subject.Cohort="Subject Cohort",FamilyHistory="Has Family Members with PD",  
  AgeatScreening = "Age at Screening", YearsofEducation = "Years of Education",  
  #BentonScore6 = "Benton Judgment of Line Orientation Score (Visit 06 (Month 24))")  
#ptree <- rpart(YearsofEducation ~ AgeatScreening, parkinsons)  
##ptree <- rpart(Subject.Cohort ~ FamilyHistory + Handedness + Gender, data=parkinson  
s)  
#rpart.plot(ptree)
```

```
#ntree <- rpart(YearsofEducation ~ AgeatScreening + BentonScore6, parkinsons)  
#rpart.plot(ntree)
```

```
stree <- rpart(Gender ~ ., data=parkinsons)  
rpart.plot(stree)
```

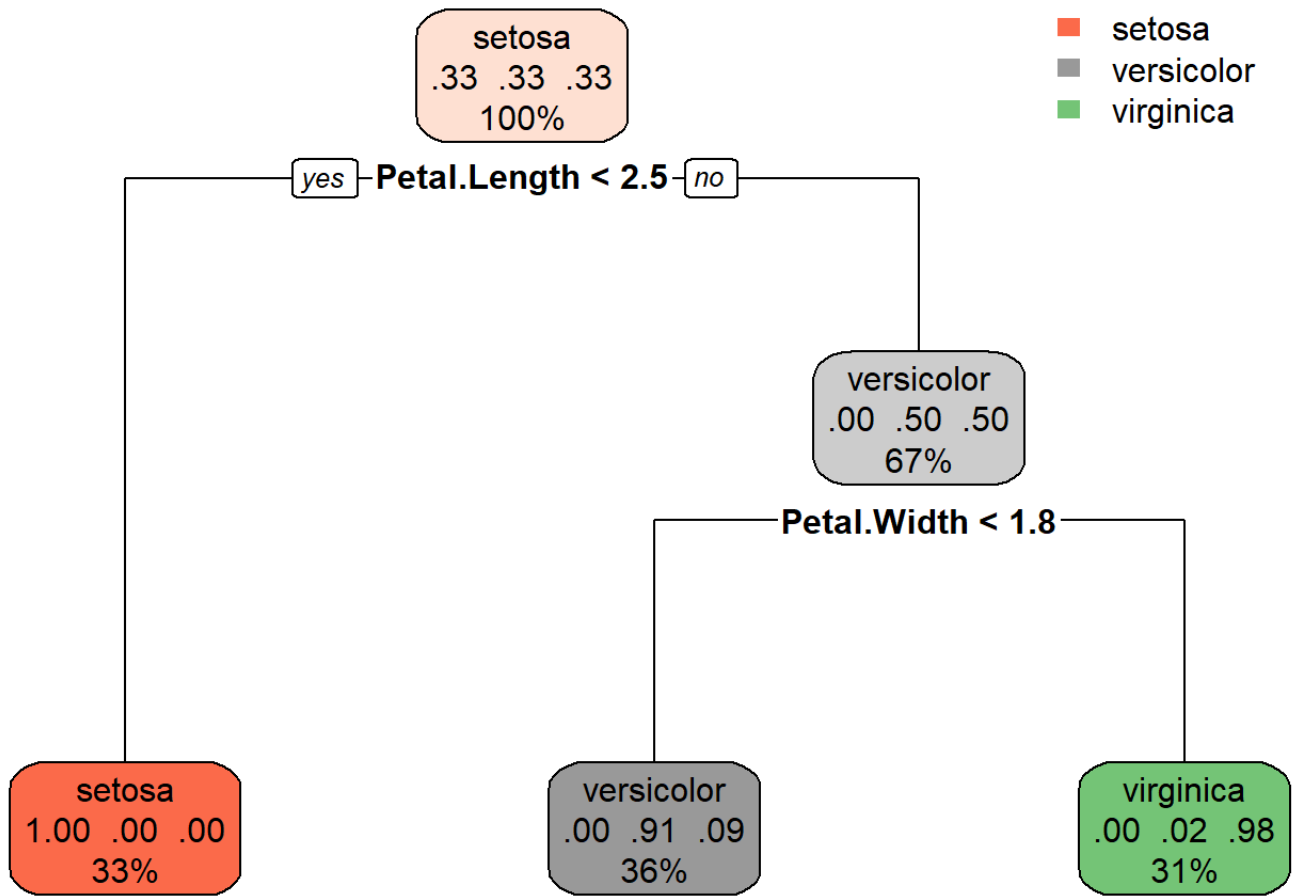




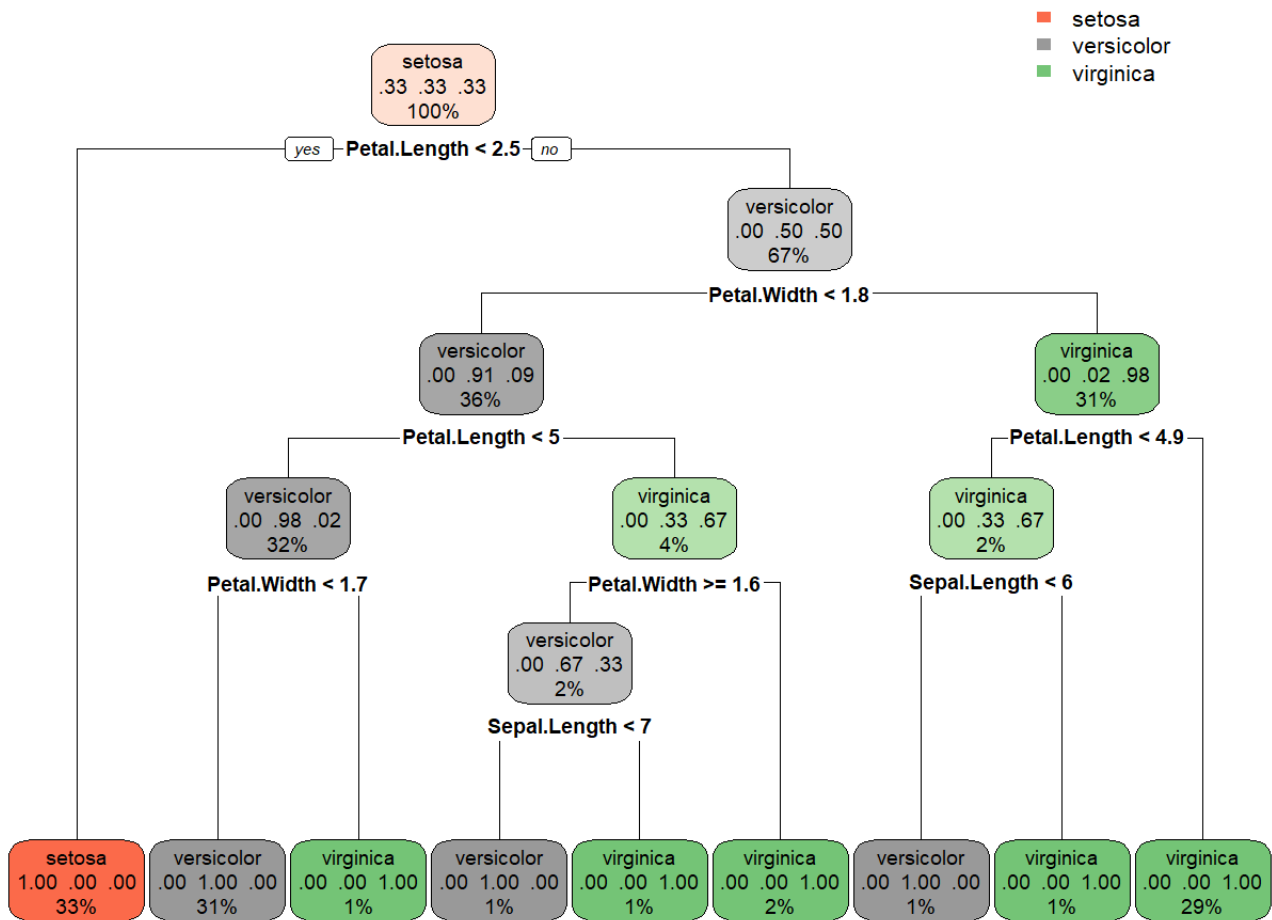
```
#s <- rpart.plot(ltree)
```

Including Plots

```
library(randomForest)
rtree1 <- rpart(Species ~.,iris)
rtree2 <- rpart(Species ~.,control = rpart.control(cp=0,minsplit=1),iris)
#complexity parameter:
#minsplit:min number of observations to be present for a split to occur
rpart.plot(rtree1)
```



```
rpart.plot(rtree2)
```



```

#set.seed(200)
#rf1 <- randomForest(Species ~., ntree = 500, mtry = 4, iris)
##number of variables randomly sampled
#rf
#plot(rf1)

```

```

library(rfUtilities)
library(randomForest)
#rf.crossValidation(rf, #iris[,c(1:4)], ydata=iris$Species, seed=590)

```

```

factor.columns <- sapply(parkinsons, function(x)
  class(x) != 'integer' &
  class(x) != 'numeric')

parkinsons[,factor.columns] <- data.frame(apply(parkinsons[,factor.columns, drop=F],
  2, as.factor), stringsAsFactors = TRUE)
parkinsons <- na.roughfix(parkinsons)

```

```

library(randomForest)
# Parkinson's data set (classification)
#data <- read.table(file = "https://archive.ics.uci.edu/ml/machine-learning-database
s/parkinsons/parkinsons.data", sep = ",", header = TRUE, check.names = FALSE)
data <- parkinsons
response <- "Subject.Cohort"

# Show data dimensions
dim(data)

```

```
## [1] 716 302
```

```
## [1] 195 23
```

```

# To demonstrate missing imputation, here we randomly select 100 values and convert t
hem to missing values
#set.seed(1) # For reproducibility
#for (i in seq_len(100)) {
#  data[sample(1:nrow(data), 1), sample(1:ncol(data), 1)] <- NA
#}

# Show number of NAs in response variable
table(data[response], useNA='ifany')

```

```

## Subject.Cohort
## Control      PD      SWEDD
##      213      422      81

```

```

## status
## 0      1 <NA>
## 48  144      3

```

```

# Remove samples with missing values in the response variable
response_var <- data[!is.na(data[response]), colnames(data) == response]
predictor_vars <- data[!is.na(data[response]), colnames(data) != response]

# Impute missing values in predictor variables
#predictor_vars <- na.roughfix(predictor_vars)

# Show dimensions of response and predictor variables
length(response_var)

```

```
## [1] 716
```

```
nrow(data)
```

```
## [1] 716
```

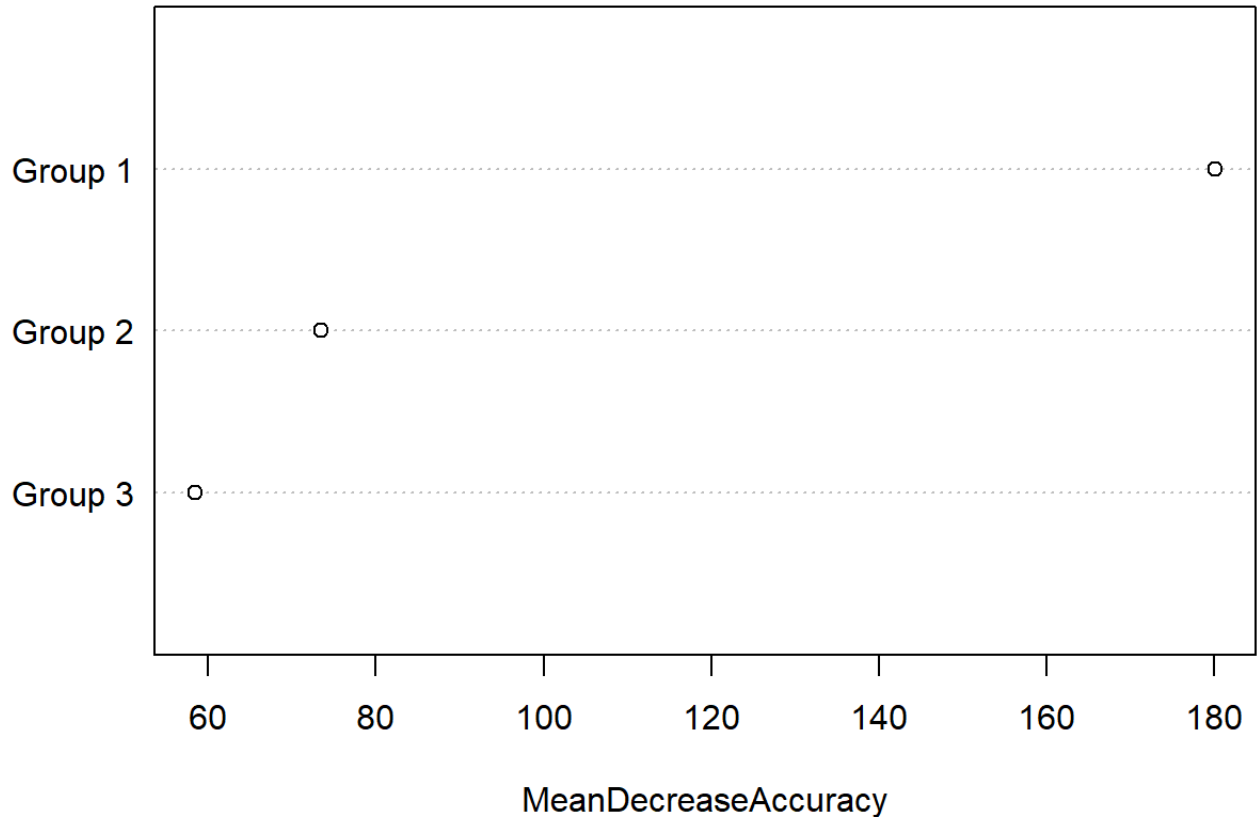
```
nrow(parkinsons)
```

```
## [1] 716
```

```
#install.packages('rfExtra_0.0.0.9000.tar.gz',type='source', #repos=NULL, force=TRUE)  
library(rfExtra)  
library(parallel)
```

```
# Set seed for reproducible results  
set.seed(1)  
  
# Downsample to the smaller class size  
response_var = as.factor(response_var)  
min_size = min(table(response_var))  
num_classes = length(table(response_var))  
  
# Train random forest  
rf = rfExtra::randomForest(x = predictor_vars, y = response_var,  
  importance = TRUE, groupImp = TRUE, ntree = 1e+05,  
  sampsize = rep(min_size, num_classes), varGroups = factor(c(rep("Group 1",  
    5), rep("Group 2", 11), rep("Group 3", 6))))  
  
# Plot group variable importance  
grpVarImpPlot(rf)
```


rf



```
impo<- data.frame(importance(rf))
impo2 <- impo[order(-impo$MeanDecreaseAccuracy),]
```

rf

```
##
## Call:
## randomForest(x = predictor_vars, y = response_var, ntree = 1e+05,      sampsize =
rep(min_size, num_classes), importance = TRUE,      groupImp = TRUE, varGroups = fact
or(c(rep("Group 1", 5),      rep("Group 2", 11), rep("Group 3", 6))))
##
##           Type of random forest: classification
##
##           Number of trees: 1e+05
## No. of variables tried at each split: 17
##
##           OOB estimate of  error rate: 0.98%
## Confusion matrix:
##           Control  PD SWEDD class.error
## Control      213   0     0 0.00000000
## PD            0 417     5 0.01184834
## SWEDD         0   2    79 0.02469136
```

```
rf$confusion
```

```
##          Control  PD  SWEDD  class.error
## Control      213    0      0  0.00000000
## PD           0  417      5  0.01184834
## SWEDD         0    2     79  0.02469136
```

```
es<- data.frame(importance(rf))
es2 <- es[order(-es$MeanDecreaseAccuracy),]
listt <- row.names(es2[0:30,])
listt
```

```
## [1] "Population.Number"
## [2] "Right.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [3] "Left.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [4] "Hoehn...Yahr.Pre.Dose..Baseline."
## [5] "MDS.UPDRS.Part.III.Pre.Dose..Baseline."
## [6] "MDS.UPDRS.Total.Pre.Dose..Baseline."
## [7] "MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit."
## [8] "Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [9] "MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36.."
## [10] "Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [11] "MDS.UPDRS.Part.II...Patient.questionanaire..Baseline."
## [12] "MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36.."
## [13] "MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit."
## [14] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36.."
## [15] "MDS.UPDRS.Total.Pre.Dose..Screening.Visit."
## [16] "Hoehn...Yahr.Pre.Dose..Visit.08..Month.36.."
## [17] "University.of.Pennsylvania.Smell.Identification.Test..UPSIT."
## [18] "MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48.."
## [19] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48.."
## [20] "Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [21] "Right.Caudate.Striatal.Binding.Ratio..Screening.Visit."
## [22] "Left.Caudate.Striatal.Binding.Ratio..Screening.Visit."
## [23] "Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [24] "MDS.UPDRS.Part.II...Patient.questionanaire..Screening.Visit."
## [25] "Hoehn...Yahr.Pre.Dose..Visit.10..Month.48.."
## [26] "Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [27] "MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30.."
## [28] "Hoehn...Yahr.Pre.Dose..Visit.04..Month.12.."
## [29] "Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [30] "MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12.."
```

```
library(tidyverse)
library(dplyr)
new_df <- select(parkinsons, c("Subject.Cohort",listt))
```

```
## Warning: Using an external vector in selections was deprecated in tidyselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
## # Was:
## data %>% select(listt)
##
## # Now:
## data %>% select(all_of(listt))
##
## See <https://tidyselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
library(randomForest)
# Parkinson's data set (classification)
#data <- read.table(file = "https://archive.ics.uci.edu/ml/machine-learning-database
s/parkinsons/parkinsons.data", sep = ",", header = TRUE, check.names = FALSE)
data <- new_df
response <- "Subject.Cohort"

# Show data dimensions
dim(data)
```

```
## [1] 716 31
```

```
## [1] 195 23

# To demonstrate missing imputation, here we randomly select 100 values and convert t
hem to missing values
#set.seed(1) # For reproducibility
#for (i in seq_len(100)) {
# data[sample(1:nrow(data), 1), sample(1:ncol(data), 1)] <- NA
#}

# Show number of NAs in response variable
table(data[response], useNA='ifany')
```

```
## Subject.Cohort
## Control      PD      SWEDD
##      213      422      81
```

```
## status
## 0      1 <NA>
## 48  144      3

# Remove samples with missing values in the response variable
response_var <- data[!is.na(data[response]), colnames(data) == response]
predictor_vars <- data[!is.na(data[response]), colnames(data) != response]

# Impute missing values in predictor variables
#predictor_vars <- na.roughfix(predictor_vars)

# Show dimensions of response and predictor variables
length(response_var)
```

```
## [1] 716
```

```
nrow(data)
```

```
## [1] 716
```

```
nrow(parkinsons)
```

```
## [1] 716
```

```
# Set seed for reproducible results
set.seed(1)

# Downsample to the smaller class size
response_var = as.factor(response_var)
min_size = min(table(response_var))
num_classes = length(table(response_var))
```

```
#rf$serr.rate
```

```
#SAMPLE EXAMPLE
beta0 = 1.5 #intercept
slope = 2.0
set.seed(2)
n = 500 #sample size
M = 1000 #number of times population is resampled

#normal distributions
error = rnorm(n, mean = 0, sd = 2)
X = rnorm(n, mean = 5, sd = 5)
Y = beta0 + slope*X + error

library(data.table)
```

```
##
## Attaching package: 'data.table'
```

```
## The following objects are masked from 'package:lubridate':
##
##      hour, isoweek, mday, minute, month, quarter, second, wday, week,
##      yday, year
```

```
## The following objects are masked from 'package:dplyr':
##
##      between, first, last
```

```
## The following object is masked from 'package:purrr':
##
##      transpose
```

```
pacman::p_load(data.table,fixest)
data = data.table(X=X,Y=Y)

#show regressions and compare to true parameters
regression <- feols(data = data, Y ~ X)
#calc conf interv
conf95 <- c(regression$coefficients[2] - 1.96 * regression$se[2],regression$coefficients[2] + 1.96 * regression$se[2])
regression
```

```
## OLS estimation, Dep. Var.: Y
## Observations: 500
## Standard-errors: IID
##           Estimate Std. Error  t value  Pr(>|t|)
## (Intercept)  1.70593    0.134954  12.6408 < 2.2e-16 ***
## X            1.98446    0.018524  107.1279 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 2.06123   Adj. R2: 0.958328
```

```
conf95 # pop parameter between these two values 95% of the time
```

```
##           X           X
## 1.948152  2.020767
```

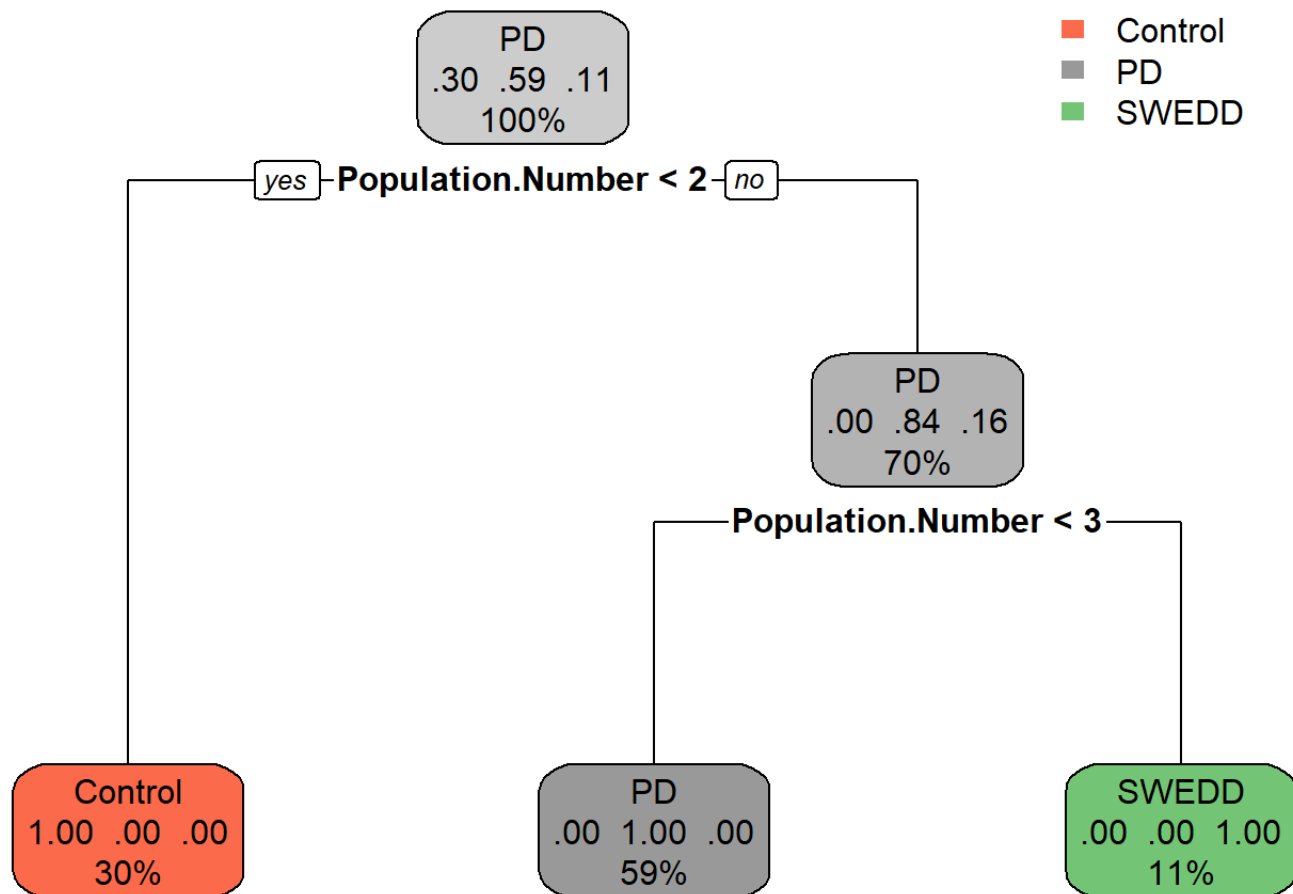
```
#bootstrap can help get an estimate of se of the slope aka:
regression$se[2]
```

```
##           X
## 0.01852422
```

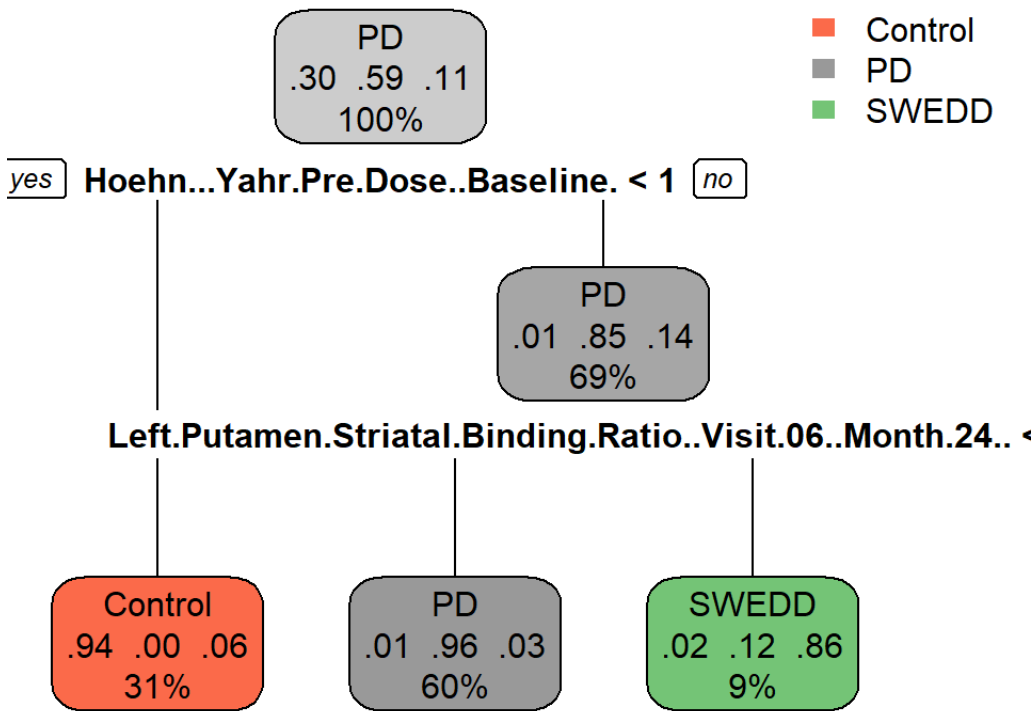
```
#loading data
parkinsons <- read_csv("ppmi_data.csv", show_col_types = FALSE)

parkinsons = data.frame(parkinsons, stringsAsFactors = TRUE)
#still has NA slots at this point
```

```
single_tree <- rpart(Subject.Cohort ~ ., parkinsons)
rpart.plot(single_tree)
```



```
parkinsons <- parkinsons[-c(1)]
single_tree <- rpart(Subject.Cohort ~ ., parkinsons)
rpart.plot(single_tree)
```



```
printcp(single_tree)
```

```
##
## Classification tree:
## rpart(formula = Subject.Cohort ~ ., data = parkinsons)
##
## Variables actually used in tree construction:
## [1] Hoehn...Yahr.Pre.Dose..Baseline.
## [2] Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..
##
## Root node error: 294/716 = 0.41061
##
## n= 716
##
##      CP nsplit rel error  xerror    xstd
## 1 0.70748      0  1.00000 1.00000 0.044774
## 2 0.16327      1  0.29252 0.31973 0.030737
## 3 0.01000      2  0.12925 0.16327 0.022762
```



```

factor.columns <- sapply(parkinsons,function(x)
  class(x) != 'integer' &
  class(x) != 'numeric')

parkinsons[,factor.columns] <- data.frame(apply(parkinsons[,factor.columns,drop=F],
2, as.factor),stringsAsFactors = TRUE)
parkinsons <- na.roughfix(parkinsons)
#Replace all numeric values that are represented by NA of the mean of that column

```

```

data <- parkinsons
response <- "Subject.Cohort"

# Show data dimensions
dim(data)

```

```
## [1] 716 301
```

```

## [1] 195 23

# To demonstrate missing imputation, here we randomly select 100 values and convert them to missing values
#set.seed(1) # For reproducibility
#for (i in seq_len(100)) {
  # data[sample(1:nrow(data), 1), sample(1:ncol(data), 1)] <- NA
#}

# Show number of NAs in response variable
table(data[response], useNA='ifany')

```

```

## Subject.Cohort
## Control      PD      SWEDD
##      213      422      81

```

```

## status
## 0      1 <NA>
## 48  144      3

# Remove samples with missing values in the response variable
response_var <- data[!is.na(data[response]), colnames(data) == response]
predictor_vars <- data[!is.na(data[response]), colnames(data) != response]

# Impute missing values in predictor variables
#predictor_vars <- na.roughfix(predictor_vars)

# Show dimensions of response and predictor variables
length(response_var)

```

```
## [1] 716
```

```
nrow(data)
```

```
## [1] 716
```

```
nrow(parkinsons)
```

```
## [1] 716
```

```
# Set seed for reproducible results
#set.seed(1)

# Downsample to the smaller class size
response_var = as.factor(response_var)
min_size = min(table(response_var))
num_classes = length(table(response_var))

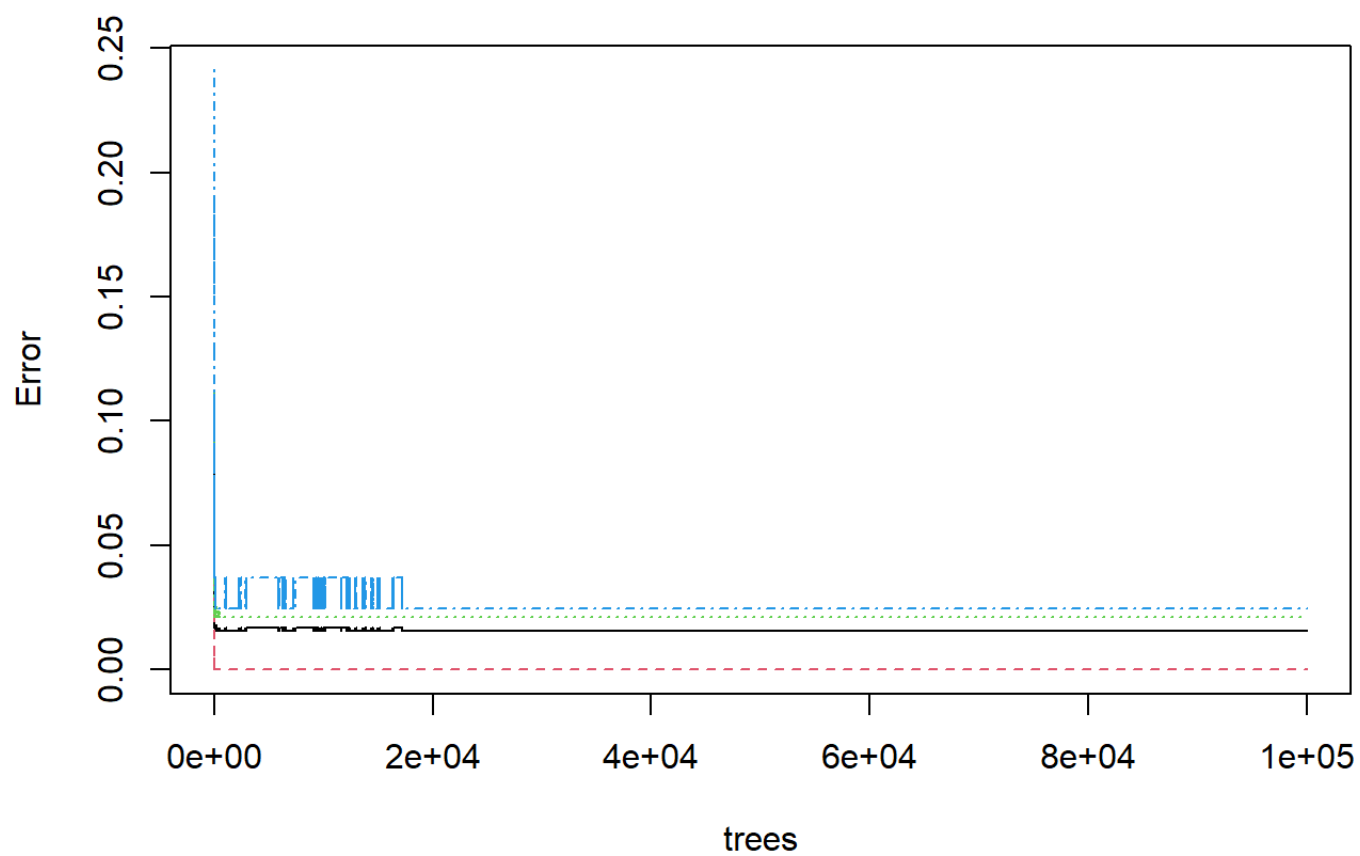
# Train random forest
#rf = rfExtra::randomForest(x = predictor_vars, y = response_var,
  #importance = TRUE, ntree = 1e+05,
  #sampsize = rep(min_size, num_classes))

# Plot group variable importance
#grpVarImpPlot(rf)
```

```
#normal rf
RF_Model = rfExtra::randomForest(x = predictor_vars, y = response_var,
  importance = TRUE, ntree = 1e+05,
  sampsize = rep(min_size, num_classes))
```

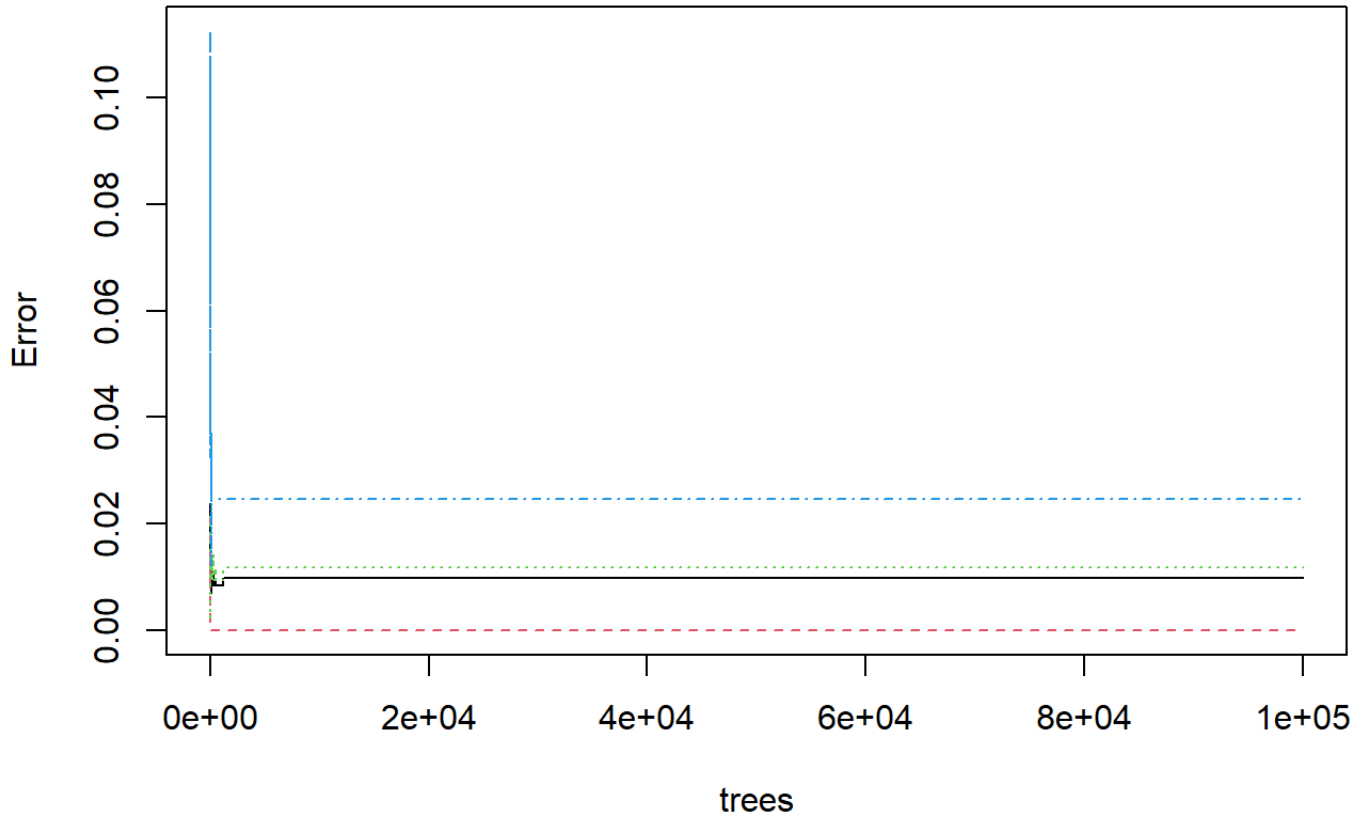
```
plot(RF_Model)
```

RF_Model



```
plot(rf)
```

rf



```
orderofimp<- data.frame(importance(rf))
orderofimp2 <- orderofimp[order(-orderofimp$MeanDecreaseAccuracy),]
show <- row.names(orderofimp2[0:20,])
show
```

```
## [1] "Population.Number"
## [2] "Right.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [3] "Left.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [4] "Hoehn...Yahr.Pre.Dose..Baseline."
## [5] "MDS.UPDRS.Part.III.Pre.Dose..Baseline."
## [6] "MDS.UPDRS.Total.Pre.Dose..Baseline."
## [7] "MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit."
## [8] "Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [9] "MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36.."
## [10] "Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [11] "MDS.UPDRS.Part.II...Patient.questionanaire..Baseline."
## [12] "MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36.."
## [13] "MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit."
## [14] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36.."
## [15] "MDS.UPDRS.Total.Pre.Dose..Screening.Visit."
## [16] "Hoehn...Yahr.Pre.Dose..Visit.08..Month.36.."
## [17] "University.of.Pennsylvania.Smell.Identification.Test..UPSIT."
## [18] "MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48.."
## [19] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48.."
## [20] "Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12.."
```

```
#orderofimp2
```

```
es<- data.frame(importance(rf))
es2 <- es[order(-es$MeanDecreaseAccuracy),]
listt <- row.names(es2[0:30,])
listt
```

```
## [1] "Population.Number"
## [2] "Right.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [3] "Left.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [4] "Hoehn...Yahr.Pre.Dose..Baseline."
## [5] "MDS.UPDRS.Part.III.Pre.Dose..Baseline."
## [6] "MDS.UPDRS.Total.Pre.Dose..Baseline."
## [7] "MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit."
## [8] "Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [9] "MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36.."
## [10] "Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [11] "MDS.UPDRS.Part.II...Patient.questionanaire..Baseline."
## [12] "MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36.."
## [13] "MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit."
## [14] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36.."
## [15] "MDS.UPDRS.Total.Pre.Dose..Screening.Visit."
## [16] "Hoehn...Yahr.Pre.Dose..Visit.08..Month.36.."
## [17] "University.of.Pennsylvania.Smell.Identification.Test..UPSIT."
## [18] "MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48.."
## [19] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48.."
## [20] "Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [21] "Right.Caudate.Striatal.Binding.Ratio..Screening.Visit."
## [22] "Left.Caudate.Striatal.Binding.Ratio..Screening.Visit."
## [23] "Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [24] "MDS.UPDRS.Part.II...Patient.questionanaire..Screening.Visit."
## [25] "Hoehn...Yahr.Pre.Dose..Visit.10..Month.48.."
## [26] "Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [27] "MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30.."
## [28] "Hoehn...Yahr.Pre.Dose..Visit.04..Month.12.."
## [29] "Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [30] "MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12.."
```

```
new_data <- new_df
new_response <- "Subject.Cohort"
```

```
# Show data dimensions
dim(new_data)
```

```
## [1] 716 31
```

```
## [1] 195 23
```

```
# To demonstrate missing imputation, here we randomly select 100 values and convert them to missing values
```

```
#set.seed(1) # For reproducibility
```

```
#for (i in seq_len(100)) {
```

```
# data[sample(1:nrow(data), 1), sample(1:ncol(data), 1)] <- NA
```

```
#}
```

```
# Show number of NAs in response variable
```

```
table(new_data[new_response], useNA='ifany')
```

```
## Subject.Cohort
```

```
## Control      PD      SWEDD
```

```
##      213      422      81
```

```
## status
```

```
## 0      1 <NA>
```

```
## 48     144     3
```

```
# Remove samples with missing values in the response variable
```

```
new_response_var <- new_data[!is.na(new_data[new_response]), colnames(new_data) == new_response]
```

```
new_predictor_vars <- new_data[!is.na(new_data[new_response]), colnames(new_data) != new_response]
```

```
# Impute missing values in predictor variables
```

```
#predictor_vars <- na.roughfix(predictor_vars)
```

```
# Show dimensions of response and predictor variables
```

```
length(new_response_var)
```

```
## [1] 716
```

```
nrow(new_data)
```

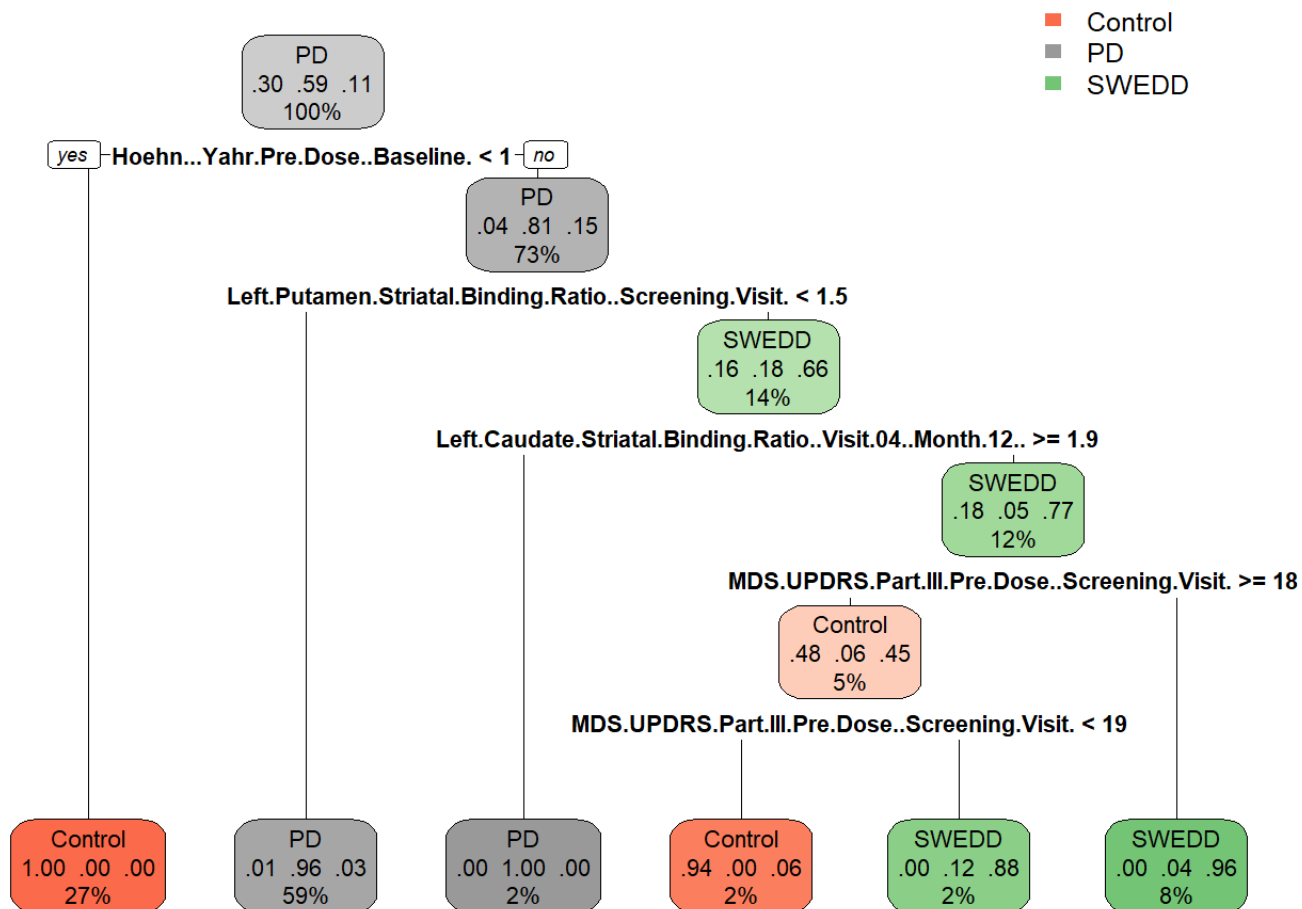
```
## [1] 716
```

```
nrow(parkinsons)
```

```
## [1] 716
```

```
single_tree <- rpart(Subject.Cohort ~ ., parkinsons)
```

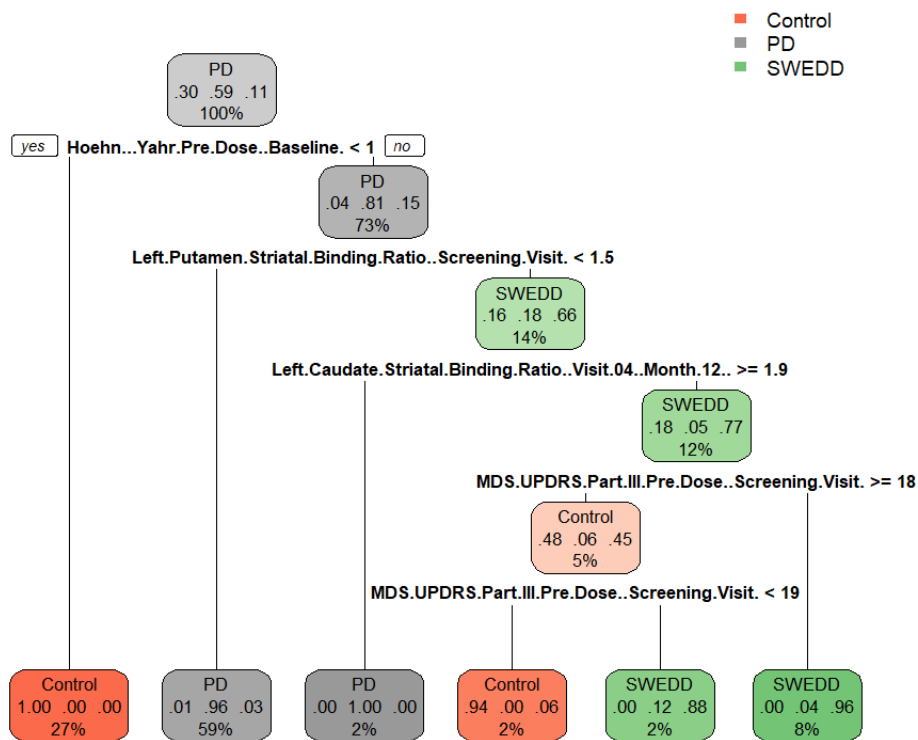
```
rpart.plot(single_tree)
```



```

parkinsons <- parkinsons[-c(1)]
single_tree <- rpart(Subject.Cohort ~ ., parkinsons)
rpart.plot(single_tree)

```

```
printcp(single_tree)
```

```
##
## Classification tree:
## rpart(formula = Subject.Cohort ~ ., data = parkinsons)
##
## Variables actually used in tree construction:
## [1] Hoehn...Yahr.Pre.Dose..Baseline.
## [2] Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..
## [3] Left.Putamen.Striatal.Binding.Ratio..Screening.Visit.
## [4] MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit.
##
## Root node error: 294/716 = 0.41061
##
## n= 716
##
##      CP nsplit rel error  xerror   xstd
## 1 0.656463      0 1.000000 1.00000 0.044774
## 2 0.166667      1 0.343537 0.34354 0.031681
## 3 0.047619      2 0.176871 0.18707 0.024237
## 4 0.025510      3 0.129252 0.16327 0.022762
## 5 0.010000      5 0.078231 0.10204 0.018236
```

```

factor.columns <- sapply(parkinsons,function(x)
  class(x) != 'integer' &
  class(x) != 'numeric')

parkinsons[,factor.columns] <- data.frame(apply(parkinsons[,factor.columns,drop=F],
2, as.factor),stringsAsFactors = TRUE)
parkinsons <- na.roughfix(parkinsons)
#Replace all numeric values that are represented by NA of the mean of that column

```

```

data <- parkinsons
response <- "Subject.Cohort"

# Show data dimensions
dim(data)

```

```
## [1] 716 300
```

```

## [1] 195 23

# To demonstrate missing imputation, here we randomly select 100 values and convert them to missing values
#set.seed(1) # For reproducibility
#for (i in seq_len(100)) {
  # data[sample(1:nrow(data), 1), sample(1:ncol(data), 1)] <- NA
#}

# Show number of NAs in response variable
table(data[response], useNA='ifany')

```

```

## Subject.Cohort
## Control      PD      SWEDD
##      213      422      81

```

```

## status
## 0      1 <NA>
## 48  144      3

# Remove samples with missing values in the response variable
response_var <- data[!is.na(data[response]), colnames(data) == response]
predictor_vars <- data[!is.na(data[response]), colnames(data) != response]

# Impute missing values in predictor variables
#predictor_vars <- na.roughfix(predictor_vars)

# Show dimensions of response and predictor variables
length(response_var)

```

```
## [1] 716
```

```
nrow(data)
```

```
## [1] 716
```

```
nrow(parkinsons)
```

```
## [1] 716
```

```
# Set seed for reproducible results
#set.seed(1)

# Downsample to the smaller class size
response_var = as.factor(response_var)
min_size = min(table(response_var))
num_classes = length(table(response_var))

# Train random forest
#rf = rfExtra::randomForest(x = predictor_vars, y = response_var,
  #importance = TRUE, ntree = 1e+05,
  #sampsize = rep(min_size, num_classes))

# Plot group variable importance
#grpVarImpPlot(rf)
```

```
#normal rf
RF_Model = rfExtra::randomForest(x = predictor_vars, y = response_var,
  importance = TRUE, ntree = 1e+05,
  sampsize = rep(min_size, num_classes))
```

```
orderofimp<- data.frame(importance(rf))
orderofimp2 <- orderofimp[order(-orderofimp$MeanDecreaseAccuracy),]
show <- row.names(orderofimp2[0:20,])
show
```

```
## [1] "Population.Number"
## [2] "Right.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [3] "Left.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [4] "Hoehn...Yahr.Pre.Dose..Baseline."
## [5] "MDS.UPDRS.Part.III.Pre.Dose..Baseline."
## [6] "MDS.UPDRS.Total.Pre.Dose..Baseline."
## [7] "MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit."
## [8] "Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [9] "MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36.."
## [10] "Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [11] "MDS.UPDRS.Part.II...Patient.questionanaire..Baseline."
## [12] "MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36.."
## [13] "MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit."
## [14] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36.."
## [15] "MDS.UPDRS.Total.Pre.Dose..Screening.Visit."
## [16] "Hoehn...Yahr.Pre.Dose..Visit.08..Month.36.."
## [17] "University.of.Pennsylvania.Smell.Identification.Test..UPSIT."
## [18] "MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48.."
## [19] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48.."
## [20] "Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12.."
```

```
#orderofimp2
```

```
es<- data.frame(importance(rf))
es2 <- es[order(-es$MeanDecreaseAccuracy),]
listt <- row.names(es2[0:30,])
listt
```

```
## [1] "Population.Number"
## [2] "Right.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [3] "Left.Putamen.Striatal.Binding.Ratio..Screening.Visit."
## [4] "Hoehn...Yahr.Pre.Dose..Baseline."
## [5] "MDS.UPDRS.Part.III.Pre.Dose..Baseline."
## [6] "MDS.UPDRS.Total.Pre.Dose..Baseline."
## [7] "MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit."
## [8] "Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [9] "MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36.."
## [10] "Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24.."
## [11] "MDS.UPDRS.Part.II...Patient.questionanaire..Baseline."
## [12] "MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36.."
## [13] "MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit."
## [14] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36.."
## [15] "MDS.UPDRS.Total.Pre.Dose..Screening.Visit."
## [16] "Hoehn...Yahr.Pre.Dose..Visit.08..Month.36.."
## [17] "University.of.Pennsylvania.Smell.Identification.Test..UPSIT."
## [18] "MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48.."
## [19] "MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48.."
## [20] "Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [21] "Right.Caudate.Striatal.Binding.Ratio..Screening.Visit."
## [22] "Left.Caudate.Striatal.Binding.Ratio..Screening.Visit."
## [23] "Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [24] "MDS.UPDRS.Part.II...Patient.questionanaire..Screening.Visit."
## [25] "Hoehn...Yahr.Pre.Dose..Visit.10..Month.48.."
## [26] "Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [27] "MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30.."
## [28] "Hoehn...Yahr.Pre.Dose..Visit.04..Month.12.."
## [29] "Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12.."
## [30] "MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12.."
```

```
new_data <- new_df
new_response <- "Subject.Cohort"
```

```
# Show data dimensions
dim(new_data)
```

```
## [1] 716 31
```

```
## [1] 195 23
```

```
# To demonstrate missing imputation, here we randomly select 100 values and convert them to missing values
```

```
#set.seed(1) # For reproducibility
```

```
#for (i in seq_len(100)) {
```

```
  # data[sample(1:nrow(data), 1), sample(1:ncol(data), 1)] <- NA
```

```
#}
```

```
# Show number of NAs in response variable
```

```
table(new_data[new_response], useNA='ifany')
```

```
## Subject.Cohort
```

```
## Control      PD      SWEDD
```

```
##      213      422      81
```

```
## status
```

```
## 0      1 <NA>
```

```
## 48     144     3
```

```
# Remove samples with missing values in the response variable
```

```
new_response_var <- new_data[!is.na(new_data[new_response]), colnames(new_data) == new_response]
```

```
new_predictor_vars <- new_data[!is.na(new_data[new_response]), colnames(new_data) != new_response]
```

```
# Impute missing values in predictor variables
```

```
#predictor_vars <- na.roughfix(predictor_vars)
```

```
# Show dimensions of response and predictor variables
```

```
length(new_response_var)
```

```
## [1] 716
```

```
nrow(new_data)
```

```
## [1] 716
```

```
nrow(parkinsons)
```

```
## [1] 716
```

```
set.seed(1)
```

```
# Downsample to the smaller class size  
new_response_var = as.factor(new_response_var)  
new_min_size = min(table(new_response_var))  
new_num_classes = length(table(new_response_var))
```

```
Top_20_Vars = rfExtra::randomForest(x = new_predictor_vars, y = new_response_var, importance = TRUE, ntree = 1e+05, new_sampsize = rep(new_min_size, new_num_classes))
```

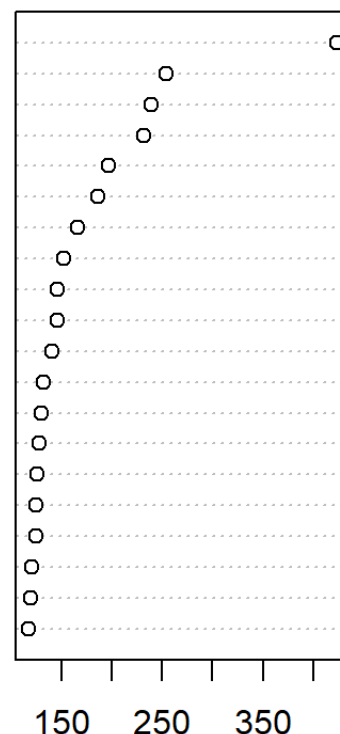
```
Top_20_Vars
```

```
##  
## Call:  
## randomForest(x = new_predictor_vars, y = new_response_var, ntree = 1e+05, importance = TRUE, new_sampsize = rep(new_min_size, new_num_classes))  
##  
##           Type of random forest: classification  
##           Number of trees: 1e+05  
## No. of variables tried at each split: 5  
##  
##           OOB estimate of error rate: 0%  
## Confusion matrix:  
##           Control  PD  SWEDD  class.error  
## Control      213    0      0           0  
## PD            0  422      0           0  
## SWEDD         0    0     81           0
```

```
varImpPlot(Top_20_Vars, n.var=20, type=1)
```

Top_20_Vars

Population.Number
 Hoehn...Yahr.Pre.Dose..Baseline.
 Left.Putamen.Striatal.Binding.Ratio..Screening.Visit.
 Right.Putamen.Striatal.Binding.Ratio..Screening.Visit.
 MDS.UPDRS.Part.III.Pre.Dose..Baseline.
 Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..
 Hoehn...Yahr.Pre.Dose..Visit.04..Month.12..
 Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..
 MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit.
 MDS.UPDRS.Total.Pre.Dose..Baseline.
 MDS.UPDRS.Part.II...Patient.questionanaire..Baseline.
 University.of.Pennsylvania.Smell.Identification.Test..UPSIT.
 Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..
 Hoehn...Yahr.Pre.Dose..Visit.08..Month.36..
 Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..
 MDS.UPDRS.Total.Pre.Dose..Screening.Visit.
 MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12..
 MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30..
 Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..
 Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..



MeanDecreaseAccuracy

```
implist<- data.frame(importance(Top_20_Vars))
implistincreasing <- implist[order(implist$MeanDecreaseAccuracy),]
implistincreasing
```

##	Control
## Left.Caudate.Striatal.Binding.Ratio..Screening.Visit.	39.80799
## Hoehn...Yahr.Pre.Dose..Visit.10..Month.48..	25.05937
## MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48..	38.99998
## Right.Caudate.Striatal.Binding.Ratio..Screening.Visit.	60.90226
## MDS.UPDRS.Part.II...Patient.questionanaire..Screening.Visit.	93.45589
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48..	61.26285
## MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36..	57.95179
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36..	70.55355
## MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit.	100.81257
## MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36..	51.00231
## Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	62.49480
## Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	61.45028
## MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30..	71.08259
## MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12..	45.76832
## MDS.UPDRS.Total.Pre.Dose..Screening.Visit.	102.59313
## Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	67.96372
## Hoehn...Yahr.Pre.Dose..Visit.08..Month.36..	35.38564

## Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	70.43517
## University.of.Pennsylvania.Smell.Identification.Test..UPSIT.	101.17360
## MDS.UPDRS.Part.II...Patient.questionanaire..Baseline.	83.16208
## MDS.UPDRS.Total.Pre.Dose..Baseline.	87.35139
## MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit.	117.17789
## Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	95.14468
## Hoehn...Yahr.Pre.Dose..Visit.04..Month.12..	53.83685
## Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	94.38901
## MDS.UPDRS.Part.III.Pre.Dose..Baseline.	89.98264
## Right.Putamen.Striatal.Binding.Ratio..Screening.Visit.	85.04094
## Left.Putamen.Striatal.Binding.Ratio..Screening.Visit.	107.02402
## Hoehn...Yahr.Pre.Dose..Baseline.	126.07724
## Population.Number	239.62111
##	PD
## Left.Caudate.Striatal.Binding.Ratio..Screening.Visit.	70.85271
## Hoehn...Yahr.Pre.Dose..Visit.10..Month.48..	93.72632
## MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48..	82.89653
## Right.Caudate.Striatal.Binding.Ratio..Screening.Visit.	68.06173
## MDS.UPDRS.Part.II...Patient.questionanaire..Screening.Visit.	36.13563
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48..	34.84300
## MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36..	63.61463
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36..	37.44327
## MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit.	15.72013
## MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36..	78.40219
## Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	91.74273
## Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	100.09423
## MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30..	91.35277
## MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12..	117.77731
## MDS.UPDRS.Total.Pre.Dose..Screening.Visit.	65.84011
## Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	93.68557
## Hoehn...Yahr.Pre.Dose..Visit.08..Month.36..	119.18069
## Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	102.49122
## University.of.Pennsylvania.Smell.Identification.Test..UPSIT.	87.48087
## MDS.UPDRS.Part.II...Patient.questionanaire..Baseline.	125.00921
## MDS.UPDRS.Total.Pre.Dose..Baseline.	125.28607
## MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit.	66.12481
## Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	104.70285
## Hoehn...Yahr.Pre.Dose..Visit.04..Month.12..	163.98533
## Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	148.71802
## MDS.UPDRS.Part.III.Pre.Dose..Baseline.	187.23981
## Right.Putamen.Striatal.Binding.Ratio..Screening.Visit.	197.20969
## Left.Putamen.Striatal.Binding.Ratio..Screening.Visit.	197.63268
## Hoehn...Yahr.Pre.Dose..Baseline.	242.41843
## Population.Number	357.76733
##	SWEDD
## Left.Caudate.Striatal.Binding.Ratio..Screening.Visit.	43.46340
## Hoehn...Yahr.Pre.Dose..Visit.10..Month.48..	91.40375
## MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48..	97.74889
## Right.Caudate.Striatal.Binding.Ratio..Screening.Visit.	53.82697
## MDS.UPDRS.Part.II...Patient.questionanaire..Screening.Visit.	60.97234

## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48..	99.77492
## MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36..	118.43095
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36..	104.06984
## MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit.	86.50040
## MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36..	115.26836
## Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	97.38942
## Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	97.16749
## MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30..	95.78473
## MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12..	32.96970
## MDS.UPDRS.Total.Pre.Dose..Screening.Visit.	86.42374
## Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	108.59553
## Hoehn...Yahr.Pre.Dose..Visit.08..Month.36..	115.26440
## Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	110.78968
## University.of.Pennsylvania.Smell.Identification.Test..UPSIT.	53.16072
## MDS.UPDRS.Part.II...Patient.questionanaire..Baseline.	66.91890
## MDS.UPDRS.Total.Pre.Dose..Baseline.	67.50393
## MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit.	123.49391
## Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	113.53419
## Hoehn...Yahr.Pre.Dose..Visit.04..Month.12..	50.61006
## Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	119.98173
## MDS.UPDRS.Part.III.Pre.Dose..Baseline.	59.61524
## Right.Putamen.Striatal.Binding.Ratio..Screening.Visit.	139.56855
## Left.Putamen.Striatal.Binding.Ratio..Screening.Visit.	153.63417
## Hoehn...Yahr.Pre.Dose..Baseline.	183.44792
## Population.Number	516.41927
##	MeanDecreaseAccur
acy	
## Left.Caudate.Striatal.Binding.Ratio..Screening.Visit.	88.50
983	
## Hoehn...Yahr.Pre.Dose..Visit.10..Month.48..	100.83
017	
## MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48..	101.40
928	
## Right.Caudate.Striatal.Binding.Ratio..Screening.Visit.	101.84
418	
## MDS.UPDRS.Part.II...Patient.questionanaire..Screening.Visit.	102.34
061	
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48..	105.27
722	
## MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36..	109.62
438	
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36..	109.86
274	
## MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit.	112.90
768	
## MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36..	114.04
767	
## Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	116.98
703	
## Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	119.85

426	
## MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30..	120.12
173	
## MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12..	124.30
775	
## MDS.UPDRS.Total.Pre.Dose..Screening.Visit.	125.21
980	
## Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	125.40
136	
## Hoehn...Yahr.Pre.Dose..Visit.08..Month.36..	128.40
001	
## Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	130.15
665	
## University.of.Pennsylvania.Smell.Identification.Test..UPSIT.	131.88
386	
## MDS.UPDRS.Part.II...Patient.questionanaire..Baseline.	140.69
965	
## MDS.UPDRS.Total.Pre.Dose..Baseline.	145.64
013	
## MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit.	145.97
149	
## Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	152.05
539	
## Hoehn...Yahr.Pre.Dose..Visit.04..Month.12..	166.14
103	
## Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	185.83
499	
## MDS.UPDRS.Part.III.Pre.Dose..Baseline.	196.38
418	
## Right.Putamen.Striatal.Binding.Ratio..Screening.Visit.	231.68
313	
## Left.Putamen.Striatal.Binding.Ratio..Screening.Visit.	239.21
022	
## Hoehn...Yahr.Pre.Dose..Baseline.	253.82
337	
## Population.Number	423.70
530	
##	MeanDecreaseGini
## Left.Caudate.Striatal.Binding.Ratio..Screening.Visit.	3.732961
## Hoehn...Yahr.Pre.Dose..Visit.10..Month.48..	9.006601
## MDS.UPDRS.Part.III.Pre.Dose..Visit.10..Month.48..	5.143238
## Right.Caudate.Striatal.Binding.Ratio..Screening.Visit.	3.834296
## MDS.UPDRS.Part.II...Patient.questionanaire..Screening.Visit.	2.544817
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.10..Month.48..	2.859948
## MDS.UPDRS.Total.Pre.Dose..Visit.08..Month.36..	6.201539
## MDS.UPDRS.Part.II...Patient.questionanaire..Visit.08..Month.36..	3.238236
## MDS.UPDRS.Part.I...Patient.questionanaire..Screening.Visit.	3.088550
## MDS.UPDRS.Part.III.Pre.Dose..Visit.08..Month.36..	7.417912
## Right.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	2.366916
## Left.Putamen.Striatal.Binding.Ratio..Visit.04..Month.12..	2.435424

## MDS.UPDRS.Total.Pre.Dose..Visit.07..Month.30..	2.105693
## MDS.UPDRS.Part.III.Pre.Dose..Visit.04..Month.12..	11.999550
## MDS.UPDRS.Total.Pre.Dose..Screening.Visit.	3.611345
## Right.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	2.555211
## Hoehn...Yahr.Pre.Dose..Visit.08..Month.36..	13.875898
## Left.Caudate.Striatal.Binding.Ratio..Visit.04..Month.12..	2.745282
## University.of.Pennsylvania.Smell.Identification.Test..UPSIT.	3.330060
## MDS.UPDRS.Part.II...Patient.questionanaire..Baseline.	6.293658
## MDS.UPDRS.Total.Pre.Dose..Baseline.	16.257688
## MDS.UPDRS.Part.III.Pre.Dose..Screening.Visit.	5.840070
## Right.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	7.670562
## Hoehn...Yahr.Pre.Dose..Visit.04..Month.12..	21.771112
## Left.Putamen.Striatal.Binding.Ratio..Visit.06..Month.24..	10.145307
## MDS.UPDRS.Part.III.Pre.Dose..Baseline.	29.626582
## Right.Putamen.Striatal.Binding.Ratio..Screening.Visit.	25.623987
## Left.Putamen.Striatal.Binding.Ratio..Screening.Visit.	34.960568
## Hoehn...Yahr.Pre.Dose..Baseline.	40.983722
## Population.Number	102.850424